

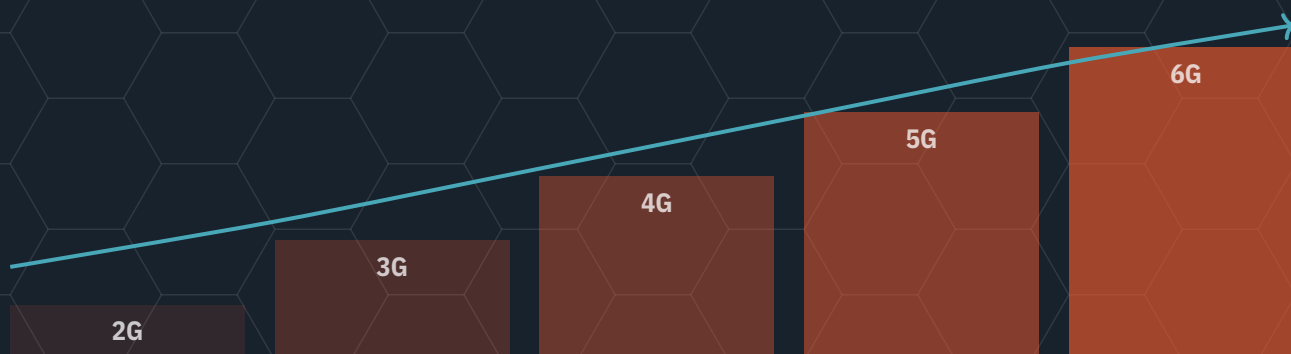
Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 25

Network Evolution — Grand Synthesis

تحول شبکه — ترکیب جامع



Lecture Notes

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1 Introduction

This module provides a **comprehensive synthesis** of mobile network evolution from 1G analog systems through 5G and beyond. It serves as a master reference connecting all previous modules, showing how each generation solved specific problems while creating new challenges that motivated the next evolution.

The evolution follows a clear trajectory:

- **Voice → Data → Everything**
- **Circuit → Packet → All-IP → Service-Based**
- **Hierarchical → Flat → Cloud-Native**
- **One-size-fits-all → Network Slicing → AI-Native**

2 Master Comparison Table

Parameter	GSM (2G)	GPRS (2.5G)	EDGE (2.75G)	UMTS (3G)
3GPP Release	Rel-96/97	Rel-97/98	Rel-99	Rel-99/4
Year	1991	2000	2003	2001
Radio Access	TDMA/FDMA	TDMA/FDMA	TDMA/FDMA (8PSK)	WCDMA
Core Network	NSS (MSC/HLR/VLR)	NSS + SGSN/GGSN	NSS + SGSN/GGSN	CS + PS Core
Switching Type	Circuit-Switched	CS + Packet-Switched	CS + PS	CS + PS
Peak DL Rate	9.6 kbps	171.2 kbps	473.6 kbps	2 Mbps
Typical Latency	500 ms	300-600 ms	200-400 ms	100-200 ms
Mobility Mechanism	Handover (hard)	Cell reselection + HO	Cell reselection + HO	Soft handover
Signaling Protocol	MAP/BSSAP	MAP/BSSGP	MAP/BSSGP	RANAP/MAP
User Plane Protocol	PCM (64 kbps)	LLC/SNDCP/GTP-U	LLC/SNDCP/GTP-U	GTP-U/PDCP
Control Plane Protocol	SS7/ISUP/MAP	SS7 + GTP-C	SS7 + GTP-C	SS7 + GTP-C
Main Innovation	Digital voice, roaming	Always-on packet data	Higher modulation (8PSK)	Wideband data (5 MHz)
Main Limitation	No data, low capacity	Slow data, shared TDMA	Still TDMA-limited	High latency, complex RNC
Subscribers (approx.)	1B (2003)	1.5B (2005)	2B (2006)	2B (2007)

Parameter	HSPA (3.5G)	LTE (4G)	LTE-Advanced (4G+)	5G NSA
3GPP Release	Rel-5/6/7	Rel-8/9	Rel-10/11/12	Rel-15
Year	2005-2008	2009	2011-2014	2018
Radio Access	WCDMA + AMC	OFDMA/SC-FDMA	OFDMA + CA + MIMO	NR + LTE
Core Network	CS + PS Core	EPC (MME/SGW/PGW)	EPC Enhanced	EPC + NGC partial
Switching Type	CS + PS (HSPA=PS)	All-IP (Packet)	All-IP	All-IP
Peak DL Rate	42 Mbps (HSPA+)	300 Mbps	3 Gbps	2-4 Gbps
Typical Latency	50-100 ms	10-30 ms	10-20 ms	4-10 ms
Mobility Mechanism	Soft/Hard HO	X2 handover (hard)	X2/S1 handover + CoMP	Dual connectivity
Signaling Protocol	RANAP/MAP	S1AP/GTP-C	S1AP/GTP-C v2	X2AP + S1AP + NGAP
User Plane Protocol	GTP-U/PDCP/MAC	GTP-U/PDCP/RLC	GTP-U/PDCP/RLC	GTP-U (dual pipe)
Control Plane Protocol	SS7 + GTP-C	Diameter/GTP-C	Diameter/GTP-C	GTP-C + NGAP
Main Innovation	Fast scheduling, AMC	All-IP, flat arch, OFDM	Carrier Aggregation, 8×8 MIMO	NR radio with LTE anchor
Main Limitation	Still CS voice, complex	No native IoT, limited MIMO	Spectrum fragmentation	Depends on 4G core
Subscribers (approx.)	3B (2012)	4B (2018)	5B (2020)	1B (2022)

Parameter	5G SA	5G Core (SBA)	6G (Future)
3GPP Release	Rel-15/16	Rel-15/16/17	Rel-20+
Year	2020	2020-2022	~2030
Radio Access	NR (standalone)	NR	Sub-THz, RIS, AI
Core Network	5GC (AMF/SMF/UPF)	5GC SBA (NFs)	6G Core (AI-native)
Switching Type	All-IP	All-IP + SBA	All-IP + Compute
Peak DL Rate	20 Gbps	20 Gbps	1 Tbps
Typical Latency	1-4 ms	1-4 ms	<0.1 ms
Mobility Mechanism	Xn handover	Xn + SBI mobility	Predictive AI HO
Signaling Protocol	NGAP	HTTP/2 + SBI	AI-native signaling
User Plane Protocol	GTP-U/SDAP	GTP-U/SDAP	Semantic comms
Control Plane Protocol	NGAP/NAS 5G	SBI (HTTP/2)	Intent-based
Main Innovation	Independent 5G, slicing	Service-Based Architecture	AI-native, sensing
Main Limitation	Coverage gaps (mmWave)	Complexity, energy	Undefined standards
Subscribers (approx.)	500M (2023)	1B+ (2024)	TBD (~2030+)

3 Generation-by-Generation Analysis

3.1 GSM (2G) — Digital Voice Revolution (1991)

Problem Solved: GSM replaced analog 1G systems (AMPS, TACS) with digital transmission, providing encryption, international roaming via SIM cards, and efficient spectrum use through TDMA. For the first time, a subscriber could travel across Europe and make calls with the same phone and number. The GSM MAP protocol enabled global mobility management, and the hierarchical NSS architecture (MSC/VLR/HLR/AuC) became the template for mobile networks.

What it couldn't do: GSM was fundamentally circuit-switched — a dedicated timeslot was allocated for the entire call duration regardless of actual data flow. Data services (CSD) were limited to 9.6 kbps, making internet access impractical. As the internet exploded in the late 1990s, operators needed packet-switched data without replacing their entire GSM infrastructure.

3.2 GPRS (2.5G) — Packet Data Overlay (2000)

Problem Solved: GPRS introduced packet-switching as an **overlay** on the existing GSM radio and core network. By adding SGSN and GGSN nodes, operators could offer “always-on” internet connectivity where users shared radio resources and were charged by data volume rather than connection time. GPRS introduced the PDP Context concept, IP addressing for mobile devices, and GTP tunneling — concepts that persist through 5G.

What it couldn't do: GPRS reused GSM's TDMA radio interface, sharing timeslots that were still primarily allocated for voice. Theoretical peak of 171.2 kbps was rarely achieved (typical: 30-50 kbps). Latency

remained high (300-600 ms) due to the TDMA frame structure. The radio interface was the bottleneck — you couldn't fix it without changing the air interface technology.

3.3 EDGE (2.75G) — Enhanced Data on GSM (2003)

Problem Solved: EDGE introduced 8PSK modulation (vs. GMSK in GSM/GPRS), tripling the bits per symbol from 1 to 3. With adaptive modulation and coding (9 MCS schemes), EDGE could deliver up to 473.6 kbps theoretical peak. It was a pure software/transceiver upgrade requiring no new infrastructure, giving operators a cost-effective capacity boost while they planned 3G deployments.

What it couldn't do: EDGE was still constrained by the 200 kHz GSM carrier bandwidth and TDMA structure. Real-world speeds of 100-200 kbps couldn't support video streaming or rich web applications. The fundamental limitation was the narrow-band radio interface — achieving true mobile broadband required a completely new wideband radio technology.

3.4 UMTS (3G) — Wideband Mobile Broadband (2001)

Problem Solved: UMTS introduced WCDMA with 5 MHz carriers, providing up to 2 Mbps peak data rates. The spread-spectrum approach offered better spectral efficiency, soft handover for improved cell-edge performance, and native support for simultaneous voice and data. The UTRAN architecture (Node B + RNC) provided centralized radio resource management, and the combined CS+PS core supported both circuit voice and packet data.

What it couldn't do: The RNC was a bottleneck and single point of failure. Latency remained high (100-200 ms round-trip) due to the complex protocol stack. The dedicated channel approach was inefficient for bursty data traffic. Voice still used circuit-switching, maintaining dual-domain complexity. The fixed spreading factor limited adaptation to varying channel conditions.

3.5 HSPA (3.5G) — High-Speed Packet Access (2005-2008)

Problem Solved: HSPA (HSDPA + HSUPA) introduced shared channels, fast scheduling at the Node B (2ms TTI), Adaptive Modulation and Coding (AMC), and HARQ — moving intelligence from the RNC to the base station. HSDPA delivered up to 14.4 Mbps (Rel-5), later HSPA+ reached 42 Mbps with 64-QAM and MIMO. This made mobile video streaming and app stores viable, launching the smartphone era.

What it couldn't do: HSPA was still built on WCDMA's 5 MHz carrier, limiting peak rates. Voice remained circuit-switched (except VoIP experiments). The RNC still existed, adding latency and cost. The architecture couldn't scale efficiently — each RNC managed hundreds of Node Bs, creating capacity planning challenges. Operators needed a clean-slate design.

3.6 LTE (4G) — All-IP Flat Architecture (2009)

Problem Solved: LTE made a revolutionary break: all-IP from day one (no circuit-switching), OFDMA for downlink (eliminating inter-cell interference issues), flat architecture (removed the RNC, eNodeB connects directly to EPC), and the Evolved Packet Core (MME/S-GW/P-GW separation). With 20 MHz bandwidth and 2×2 MIMO, LTE delivered 150-300 Mbps peak. VoLTE replaced circuit-switched voice. The X2 interface enabled direct inter-eNodeB handovers without core involvement.

What it couldn't do: LTE lacked native support for massive IoT (NB-IoT came later in Rel-13), couldn't achieve sub-millisecond latency for industrial applications, had limited MIMO (max 8 layers), and the

EPC's GTP-based point-to-point interfaces couldn't scale for the cloud-native era. Network slicing wasn't possible with monolithic EPC nodes.

3.7 LTE-Advanced (4G+) — Carrier Aggregation Era (2011-2014)

Problem Solved: LTE-Advanced introduced Carrier Aggregation (combining up to 5×20 MHz = 100 MHz), 8×8 MIMO (8 layers DL), enhanced inter-cell interference coordination (eICIC/FeICIC), relay nodes, and Coordinated Multi-Point (CoMP). Peak rates reached 3 Gbps (Rel-12). Heterogeneous Networks (HetNets) with small cells addressed capacity in dense areas. LAA (Licensed Assisted Access) added unlicensed spectrum.

What it couldn't do: LTE-A was an evolution, not revolution — it couldn't fundamentally change the frame structure for ultra-low latency, couldn't use mmWave spectrum, and the EPC remained monolithic. Supporting diverse verticals (automotive, industry, healthcare) with one network configuration was impossible. The industry needed a new radio AND a new core.

3.8 5G NSA — Non-Standalone (2018)

Problem Solved: 5G NSA (Option 3/3a/3x) deployed NR (New Radio) using the existing LTE core (EPC) as an anchor. This allowed operators to launch 5G services quickly without building an entirely new core network. Dual Connectivity (EN-DC) combined LTE and NR carriers, delivering multi-Gbps throughput using mmWave and sub-6 GHz NR spectrum while maintaining LTE for control plane and voice (VoLTE).

What it couldn't do: 5G NSA couldn't deliver the full 5G promise: no network slicing (EPC doesn't support it), no ultra-reliable low-latency (URLLC requires 5GC), no Service-Based Architecture, and the LTE anchor added latency. It was a "5G radio with 4G brain" — a necessary stepping stone, not the destination.

3.9 5G SA — Standalone (2020)

Problem Solved: 5G SA deployed NR with the 5G Core (5GC), eliminating LTE dependency. This enabled network slicing (isolated virtual networks per use case), URLLC with 1ms latency, native support for massive IoT (mMTC), edge computing (MEC) integration, and the Service-Based Architecture. The AMF/SMF/UPF separation allowed independent scaling, and the SBI (HTTP/2, JSON) enabled cloud-native deployments.

What it couldn't do: mmWave coverage remains limited (short range, blockage-sensitive), energy consumption is significantly higher than 4G, the complexity of managing network slices across operators is unsolved, and true deterministic latency for industrial automation requires further enhancements. Migration from NSA to SA requires significant investment with uncertain near-term ROI.

3.10 5G Core — Service-Based Architecture (2020-2022)

Problem Solved: The 5G Core replaced point-to-point interfaces with a Service-Based Architecture where Network Functions (NFs) expose services via RESTful APIs (HTTP/2). NRF enables service discovery, NSSF manages slice selection, PCF provides policy, and UDM unifies subscriber data. Cloud-native principles (microservices, stateless design, containerization) enable elastic scaling, multi-vendor interoperability, and rapid feature deployment.

What it couldn't do: SBA complexity challenges smaller operators. Inter-PLMN roaming with SBA (SEPP) is still maturing. Energy efficiency of distributed cloud infrastructure is a concern. True end-

to-end automation (zero-touch provisioning) requires AI/ML integration that's still evolving. The architecture is ready for 6G evolution but needs AI-native capabilities built in from the ground up.

3.11 6G (Future) — AI-Native Networks (~2030)

Problem Solved (Expected): 6G aims to deliver 1 Tbps peak rates using sub-THz frequencies (100 GHz–3 THz), sub-0.1 ms latency, integrated sensing and communication (ISAC), AI-native air interface and network management, Reconfigurable Intelligent Surfaces (RIS) for programmable propagation, non-terrestrial networks (NTN) integration for global coverage, and semantic/goal-oriented communications that transmit meaning rather than bits.

What it couldn't do (Anticipated challenges): Sub-THz propagation is extremely limited (meters to tens of meters), requiring ultra-dense deployments. Energy efficiency at these frequencies is poor. Standardization of AI-native protocols is unprecedented. Privacy concerns with pervasive sensing. Regulatory frameworks for sub-THz spectrum allocation are undefined. Commercial viability for many proposed features remains unproven.

4 Timeline Diagram

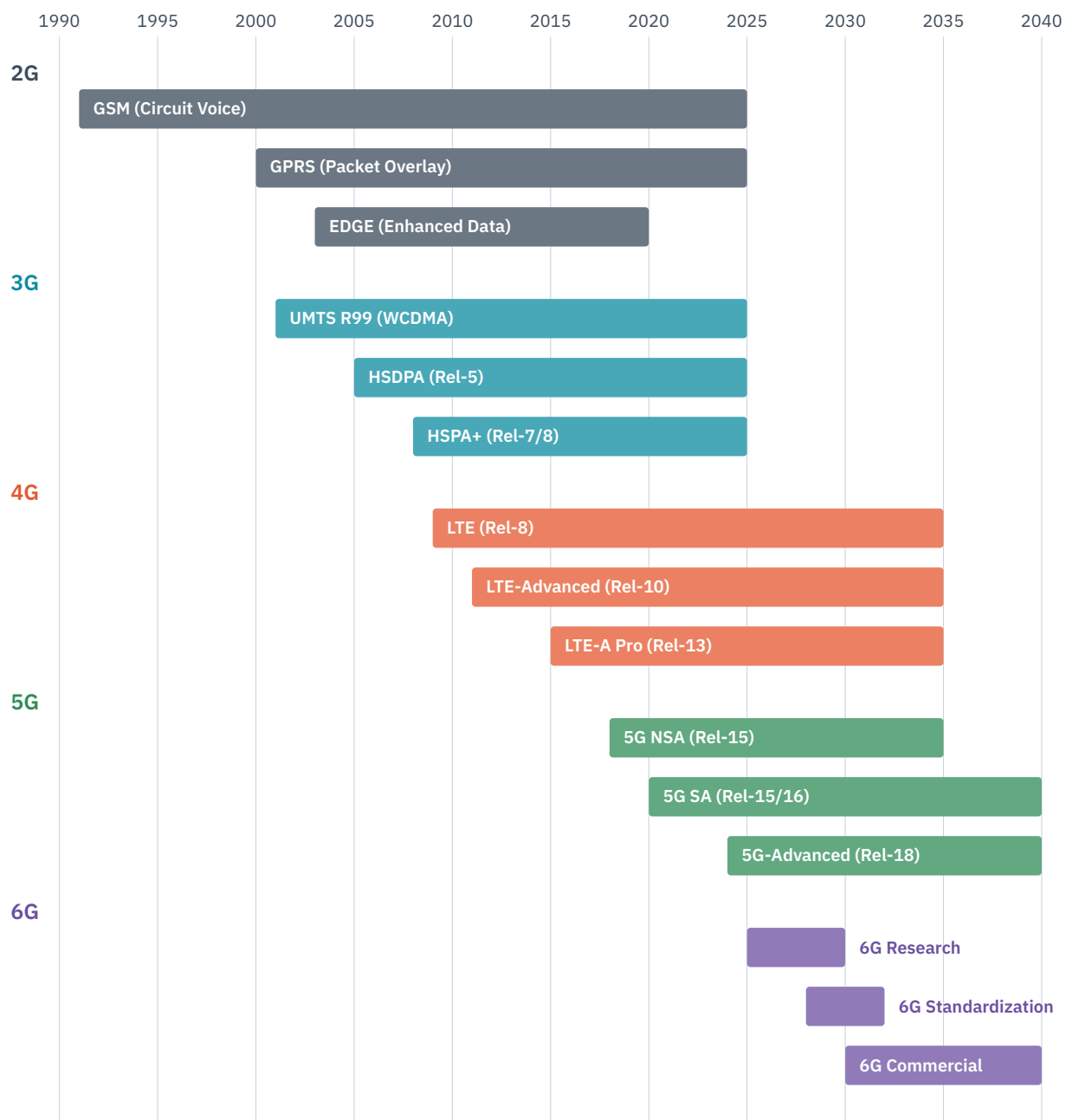


Figure 25.1 Mobile network generations on a timeline

5 Architecture Evolution Diagram

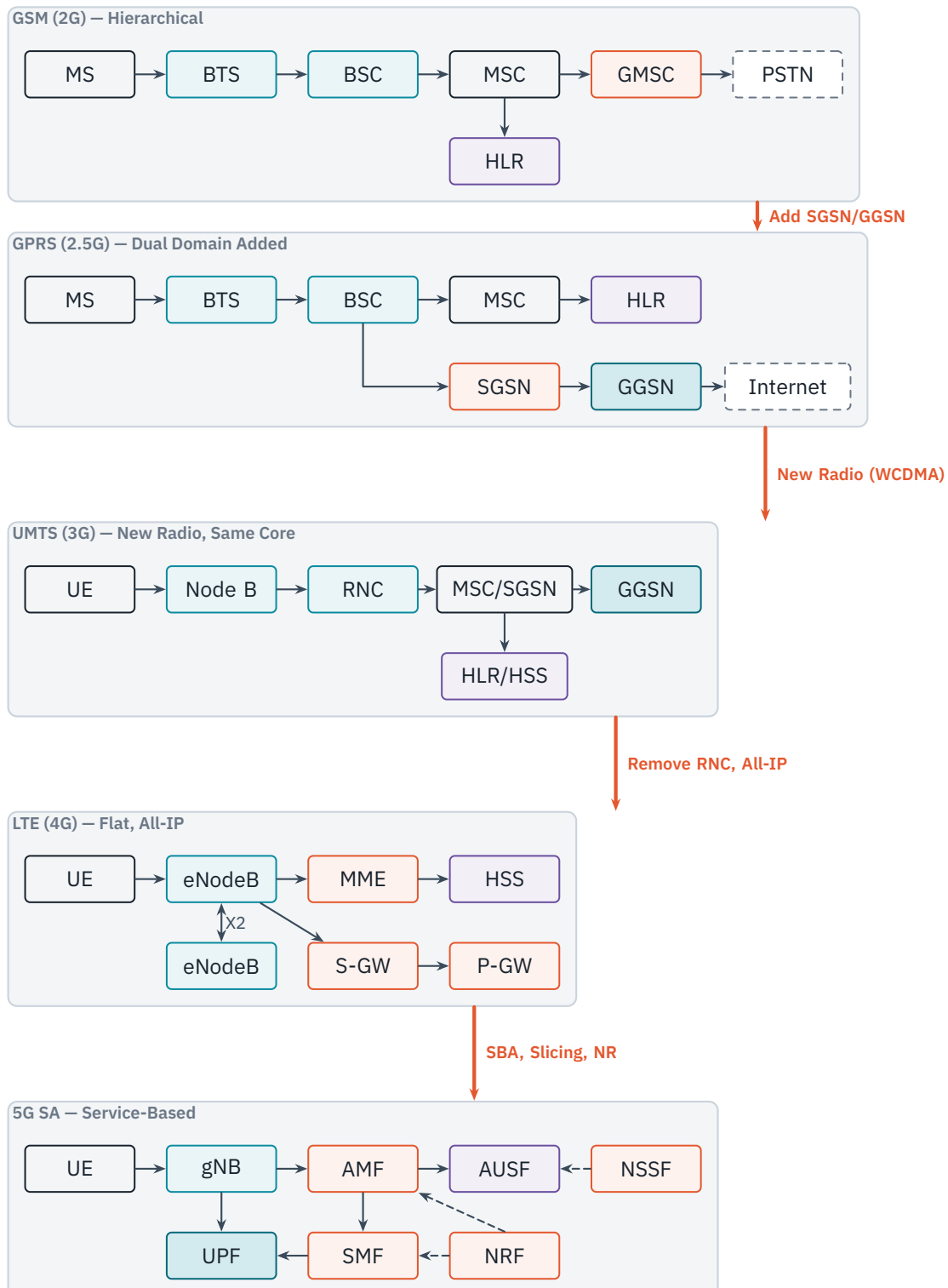


Figure 25.2 Architecture evolution from GSM to the 5G service-based core

Architecture Element Evolution Map

2G Element	3G Equivalent	4G Equivalent	5G Equivalent	Evolution Pattern
BTS	Node B	eNodeB	gNB	More intelligence pushed to edge
BSC	RNC	— (removed)	—	Eliminated (functions in eNB/gNB)
MSC	MSC Server	MME	AMF	Control plane separated from user plane
HLR	HSS	HSS	UDM/UDR	Unified data management
GMSC	GMSC	—	—	Eliminated (all-IP, no PSTN gateway needed)
VLR	VLR	MME (integrated)	AMF (stateless)	From stateful to stateless
SGSN	SGSN	S-GW (partial)	SMF+UPF	Split into control + user plane
GGSN	GGSN	P-GW	UPF	Pure user plane forwarding
—	—	PCRF	PCF	Policy and charging
—	—	—	NRF	Service discovery (new in 5G)
—	—	—	NSSF	Slice selection (new in 5G)

6 Data Rate Evolution Graph (Exponential Growth)

peak downlink data rate

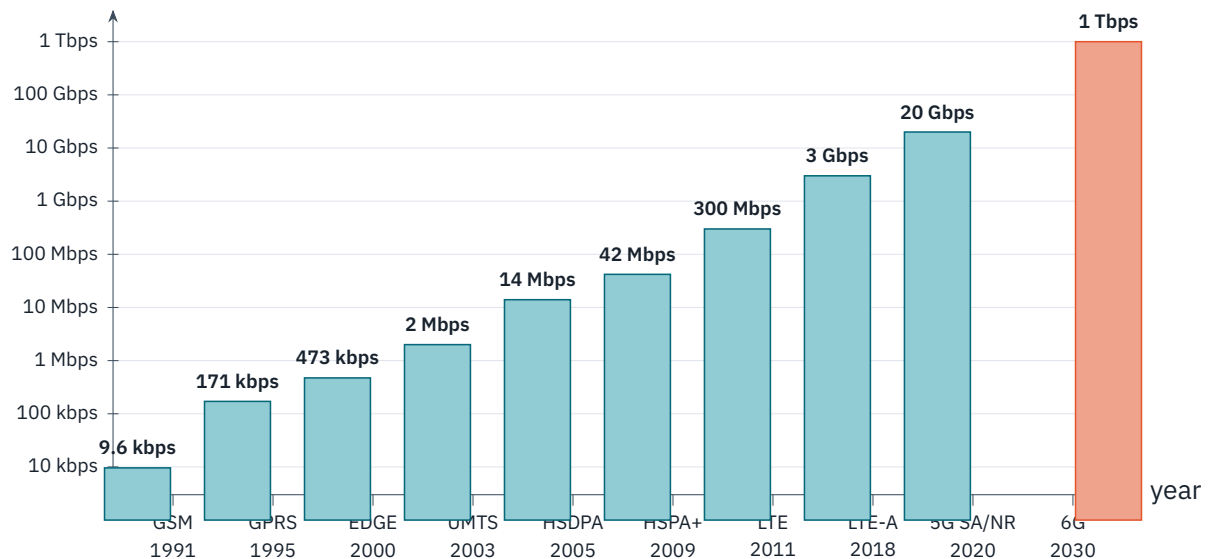


Figure 25.3 Peak downlink data rate of each generation (logarithmic scale)

Growth pattern: about $10\times$ every 5 years ($1000\times$ per decade).

Step	Improvement	Peak rate
GSM → GPRS	$18\times$	9.6 kbps → 171 kbps
GPRS → EDGE	$2.8\times$	171 kbps → 473 kbps
EDGE → UMTS	$4.2\times$	473 kbps → 2 Mbps
UMTS → HSPA	$21\times$	2 Mbps → 42 Mbps
HSPA → LTE	$7\times$	42 Mbps → 300 Mbps
LTE → LTE-A	$10\times$	300 Mbps → 3 Gbps
LTE-A → 5G	$6.7\times$	3 Gbps → 20 Gbps
5G → 6G	$50\times$	20 Gbps → 1 Tbps

7 Key Lessons Learned from Each Transition

7.1 GSM → GPRS: Overlay Packet on Circuit

Aspect	Lesson
Strategy	You don't need to replace everything — overlay new capability on existing infrastructure
Architecture	Added SGSN/GGSN alongside MSC without disrupting voice service
Risk Management	Incremental deployment: operators could add packet capability node by node
Trade-off	Shared radio resources meant neither voice nor data was optimal
Legacy Impact	GTP tunneling protocol invented here persists through 5G
Key Principle	"Evolution over revolution when the installed base is massive"

7.2 GPRS → UMTS: New Radio, Keep Core

Aspect	Lesson
Strategy	When the radio is the bottleneck, change the radio but keep the proven core
Architecture	WCDMA replaced TDMA, but SGSN/GGSN/HLR remained largely unchanged
Risk Management	Operators could run 2G and 3G simultaneously with shared core
Trade-off	New spectrum required (2100 MHz band) — massive licensing costs
Legacy Impact	RNC architecture created a new bottleneck that took a decade to remove
Key Principle	"Change one major subsystem at a time to manage risk"

7.3 UMTS → LTE: Flatten, All-IP, Remove RNC

Aspect	Lesson
Strategy	Sometimes you must break backward compatibility for fundamental improvement
Architecture	Eliminated RNC (functions distributed to eNodeB), all-IP (eliminated circuit-switching)
Risk Management	VoLTE took years to mature — interim CSFB to 2G/3G for voice
Trade-off	Clean-slate design required parallel networks for many years
Legacy Impact	Flat architecture proved far more scalable and cost-effective
Key Principle	“Flatten the hierarchy — push intelligence to the edge, simplify the core”

7.4 LTE → LTE-Advanced: Carrier Aggregation, More MIMO

Aspect	Lesson
Strategy	Combine existing resources more cleverly before seeking new ones
Architecture	CA bonds fragmented spectrum; MIMO multiplies spatial capacity
Risk Management	Backward compatible — LTE devices work alongside LTE-A
Trade-off	Complexity in scheduler and UE increases significantly
Legacy Impact	Established that spectrum aggregation is key to capacity scaling
Key Principle	“Maximize what you have through intelligent combination before building new”

7.5 LTE → 5G: New Radio + New Core, Slicing, SBA

Aspect	Lesson
Strategy	When requirements diverge (eMBB vs. URLLC vs. mMTC), one configuration cannot serve all
Architecture	NR provides flexible numerology; 5GC SBA enables per-slice customization
Risk Management	NSA deployment (Option 3x) as stepping stone before SA
Trade-off	Enormous complexity; operators must justify ROI for diverse verticals
Legacy Impact	SBA and cloud-native principles prepare for continuous software evolution
Key Principle	“Design for diversity — network slicing lets one physical network serve many purposes”

7.6 5G → 6G: AI-Native, Sensing, Sub-THz

Aspect	Lesson
Strategy	The network itself becomes intelligent — AI is not an add-on but a fundamental design principle
Architecture	Digital twin of the network, predictive resource management, semantic communications
Risk Management	Research phase allows exploration of feasibility before standardization
Trade-off	Sub-THz range is extremely limited; cost of ultra-dense deployment is unknown
Legacy Impact	May redefine what “communication” means (sensing + computing + communicating)
Key Principle	“Converge communication, computation, and sensing into a unified intelligent fabric”

8 Summary of Evolutionary Patterns

Pattern 1: Alternating Radio and Core Innovation

Generation	2G	2.5G	3G	3.5G	4G	4G+	5G
Focus	Radio	Core	Radio	Radio	Both	Radio	Both
Key innovation	TDMA	SGSN	WCDMA	HSPA	OFDMA	CA	NR + SBA

Pattern 2: Complexity Cycle

The cycle



Example

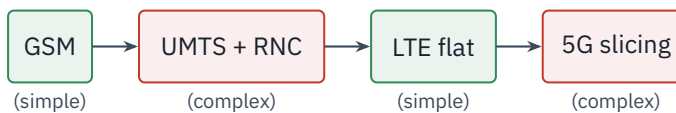


Figure 25.4 The complexity cycle of mobile network architectures

Pattern 3: Control/User Plane Separation Timeline

Generation	Control/user-plane arrangement	Result
2G	MSC handles both	Inefficient
3G	MSC-Server + MGW split (Rel-4)	Partial CUPS
4G	MME (CP) + S-GW/P-GW (UP)	Architectural CUPS
5G	AMF/SMF (CP) + UPF (UP)	Complete CUPS

Pattern 4: Where Intelligence Resides

Generation	Where the intelligence resides
2G	BSC/MSC (centralised)
3G	RNC (centralised)
4G	eNodeB (distributed to the edge)
5G	gNB + edge computing (distributed + edge cloud)
6G	Everywhere (AI at every layer)

9 Exam Preparation Quick Reference

Must-Know Facts per Generation

Gen	Remember This
GSM	TDMA, 200 kHz carrier, 8 timeslots, circuit-switched, MAP/SS7, HLR/VLR/MSC
GPRS	Added SGSN+GGSN, PDP Context, GTP tunnel, packet overlay on CS
EDGE	8PSK modulation (3 bits/symbol vs 1), 9 MCS schemes, software upgrade
UMTS	WCDMA, 5 MHz, spreading codes, RNC, soft handover, Rake receiver
HSPA	Shared channel, 2ms TTI, AMC, HARQ, fast scheduling at Node B
LTE	OFDMA, flat (no RNC), all-IP, EPC (MME/SGW/PGW), X2 interface
LTE-A	CA (5×20 MHz), 8×8 MIMO, CoMP, HetNets, Relay
5G NSA	EN-DC, LTE anchor, NR secondary, Option 3x, EPC core
5G SA	AMF/SMF/UPF, NGAP, network slicing, URLLC, mMTC
5GC	SBA, HTTP/2, NRF discovery, NSSF, stateless NFs, cloud-native
6G	Sub-THz, RIS, AI-native, ISAC, semantic comms, 1 Tbps

Key Protocol Stack Evolution

2G	3G	4G	5G
App	App	App	App
TCP/IP	TCP/IP	TCP/IP	TCP/IP
SNDCP/LLC	PDCP	PDCP	SDAP/PDCP
RLC	RLC	RLC	RLC
MAC	MAC	MAC	MAC
GSM PHY (TDMA)	WCDMA PHY (CDMA)	OFDMA PHY	NR PHY (flex. num.)

Figure 25.5 User-plane protocol stack of each generation

Critical Comparison: 4G EPC vs 5G Core

Feature	4G EPC	5G Core
Interface type	Point-to-point (GTP-C, Diameter)	Service-Based (HTTP/2 REST)
Discovery	Static configuration	NRF dynamic discovery
Scaling	Vertical (bigger boxes)	Horizontal (more instances)
Deployment	Appliance-based	Cloud-native (containers, K8s)
Slicing	Not supported	Native (NSSF, per-slice NFs)
State	Stateful nodes	Stateless NFs + external state (UDSF)
Mobility anchor	GTP tunnel (S-GW)	UPF (can be distributed)
Policy	PCRF (Diameter)	PCF (SBI)
Subscriber data	HSS (Diameter)	UDM/UDR (SBI)
Authentication	MME + HSS (EAP-AKA)	AUSF + UDM (5G-AKA, EAP)

10 Conclusion

The evolution from GSM to 6G represents one of the most remarkable engineering journeys in history — from 9.6 kbps voice calls to terabit-per-second intelligent networks in four decades. Each generation solved the previous generation’s limitations while introducing new complexity that demanded the next evolution.

The three constants across all generations:

1. **Demand always exceeds supply** — users always find ways to consume available capacity
2. **Simplification follows complexity** — every overly complex architecture is eventually flattened
3. **The boundary between network and computing blurs** — from dumb pipes to intelligent platforms

Understanding this evolution is not just historical knowledge — it teaches us how to evaluate future technologies, predict architectural decisions, and design systems that can gracefully evolve.

Module 25 — Network Evolution Grand Synthesis

Advanced Communication Networks Course

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